

Exercices Lab 2 Bonus

Frequency Analysis

1.1 Given Ciphertext

```
DLOCHVQLROC, HFHTFRZHIORZOLHVCFIFVOYHFKBOCXQIFOKHCLWOCDFKWHOELCOIFOKHXFDKFWZIHRLCELKI. HIZLHR
LCELKIHZQRHOTOSLKLDHOKDHRLLSRHFIRHECLM. HOYEZQHGPYPHIZCLLHZORHFDLKIFBFLDQHHCLOSIZCOPWZHFHKLKL
XMHYOWERIFVR. HVOALCIHORRLIRHFKHRLVIOCHHCQAOKFKLHZQALHVOKBFCXLDHIZQIHZLHRZFEXLKIHOHBLNELKRFA
LHQCXOCHTFYYHEQRRHIZCOPWZHCOPILHLVZOHBFAHIOXOCCOTHKFWZIHQRHELCHIZLHYOILRIHILYLXLCMHDOIQ. HVO
XXPKFVOIFOKHTFYHHHLKVCMEILDHTFIZHVQLROCHVFEZLCHDPCFKWHIZLHOELCOIFOK. HDLVCMEIHPRFKWHIZLHROXL
HSLMHPKOKHCLVLEI. HFHPCWLHMOPHIOHISLHIZLHPIXORIHECLVOPIFOKRHFKHZOKDYFKWHIZFRXLRRQWLHOKDHIOH
LKRPLHIZQIHZLHFKBOCXQIFOKHCLXQFKRHSKOTKHOKYMHIOHICPRILDHEL CROKKLY. HCLXLXHLCHTLHOC LHIZLHWPOC
DFQKRHOHBOPCHKQIFOKRHLVPCFIMHOKDHOPCHRPVVLRRHZEKWL RHOKHRLVCLVMHQKDHECLVFRFOK. HMOPCRHFKHCLROY
AL, HHCOAL.
```

1 Solution

This code analyzes the frequency of characters in the given ciphertext. It first removes any newline characters to process the text as a single string, then counts how many times each character appears. Finally, it displays the frequency counts sorted from most to least common, along with total character statistics.

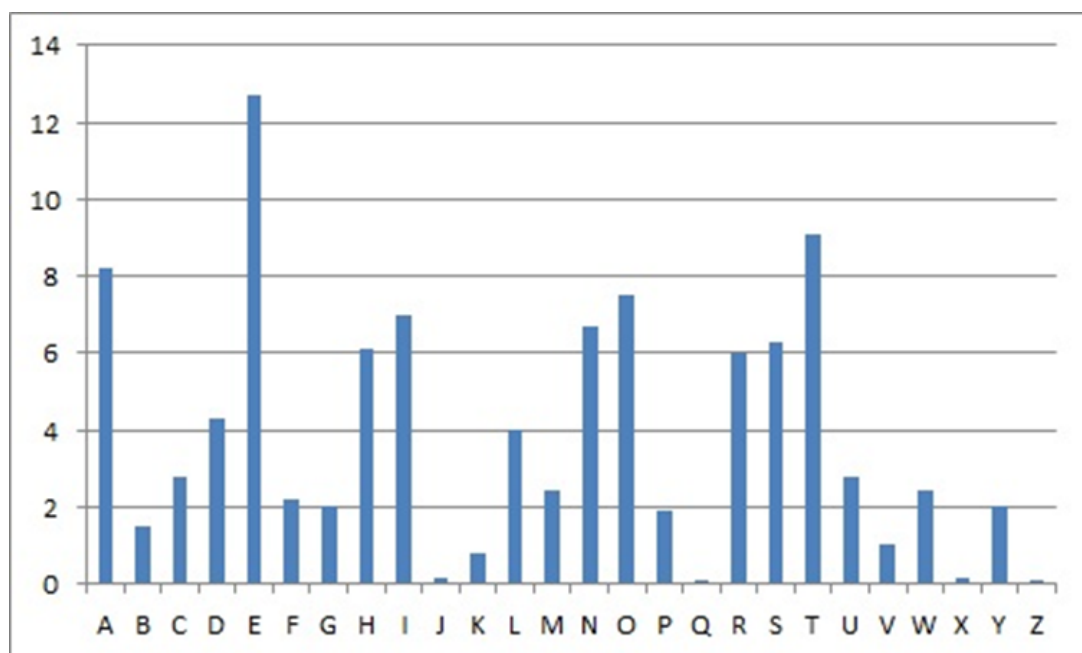
```
1 # Remove newlines for counting (ciphertext copied from slides ->
  contains newlines)
2 cleaned_text = ciphertext.replace('\n', '')
3
4 # Count frequency of each character
5 from collections import Counter
6 char_count = Counter(cleaned_text)
7
8 # Sort by frequency (descending)
9 sorted_chars = char_count.most_common()
10
11 # Print results
12 print("Character frequencies:")
13 print("-" * 30)
14 for char, count in sorted_chars:
15     print(f"'{char}': {count:3d} times")
16
17 print(f"\nTotal characters: {len(cleaned_text)}")
18 print(f"Unique characters: {len(char_count)}")
19
```

Output

Using the number of times the characters are shown, we can start the decrypting process by substituting letters.

```
Character frequencies:
-----
'H': 124 times
'L': 83 times
'I': 56 times
'C': 52 times
'F': 49 times
'K': 49 times
'Q': 47 times
'R': 45 times
'O': 43 times
'Z': 30 times
'V': 22 times
'P': 21 times
'D': 18 times
'E': 18 times
'X': 17 times
'Y': 14 times
'W': 14 times
'M': 11 times
'.' : 9 times
'T': 8 times
'B': 7 times
'A': 7 times
'S': 6 times
',' : 2 times
'G': 1 times
'N': 1 times

Total characters: 754
Unique characters: 26
```



Substitutions and Analysis

First, we substitute:

$$H \rightarrow \text{space}, \quad L \rightarrow e, \quad I \rightarrow t$$

```
DeQC VQeROC, F TFRZ to RZOCe VCftFVOY FKBOCXQtFOK CeWOCDFKW OEeCQtFOK XFDKFWZt ReCEeKt. tZe
ReCEeKt ZQR QTQSeKeD QKD ReesSR Ftr ECeM. QYEZQ GPYP tZCee ZQR FDeKtFBFeD Q CeQStZCOPWZ FK
eKeXM YOWFRtFVR. VOAcT QRRetR FK ReVtOC CQAO KfKe ZOAe VOKBFCXeD tZQt tZe RZFEXeKt OB
eNEeKRFAe QCXOC TFYY EQRR tZCOPWZ COpte eVZO BFae tOXOCCOT KFWZt QR EeC tZe YoteRt
teYeXetCM DotQ. VOXXPKFVQtFOK TFYY e eKVCMeteD TftZ VQeROC VFEZeC DPCFKW tZe OEeCQtFOK.
DeVCMet PRFKW tZe ROXe SeM PEOk CeVeFet. F PCWe MOP to tQSe tZe PtXORT ECeVQPtFOKR FK
ZQKDyFKW tZFR XerRQWe QKD to eKRPCe tZQt tZe FKBOCXQtFOK CeXQFKR SKOTK OKYM to tCPRted
EeCROKKeY. CeXeX eC Te QCe tZe WPQCDfQKR OB OPC KOfFOKR ReVPCFtM QKD OPC RPVVeRR ZFKWeR OK
ReVCeVM QKD ECeVFRFOK. MOPCR FK CeROYAe, COAe.
```

Then we identify potential patterns:

- Possibly, $F \rightarrow i$

```
DeQC VQeROC, i TirZ to RZOCe VCitivOY iKBOCXQtioK CeWOCDiKW OEeCQtioK XiDKiWZt ReCEeKt. tZe
ReCEeKt ZQR QTQSeKeD QKD ReesSR itr ECeM. QYEZQ GPYP tZCee ZQR iDeKtiBieD Q CeQStZCOPWZ iK
eKeXM YOWiRtiVR. VOAcT QRRetR iK ReVtOC CQAO KiKe ZOAe VOKBiCXeD tZQt tZe RZiEXeKt OB
eNEeKRiAe QCXOC TiYY EQRR tZCOPWZ COpte eVZO BiAe tOXOCCOT KiWZt QR EeC tZe YoteRt
teYeXetCM DotQ. VOXXPKiVQtioK TiYY e eKVCMeteD TitZ VQeROC ViEZeC DPCiKW tZe OEeCQtioK.
DeVCMet PRiKW tZe ROXe SeM PEOk CeVeiet. i PCWe MOP to tQSe tZe PtXORT ECeVQPtioKR iK
ZOKDYiKW tZiR XerRQWe QKD to eKRPCe tZQt tZe iKBOCXQtioK CeXQiKR SKOTK OKYM to tCPRted
EeCROKKeY. CeXeX eC Te QCe tZe WPOCDiOKR OB OPC KOfiOKR ReVPCitM QKD OPC RPVVeRR ZiKWeR OK
ReVCeVM QKD ECeViRiOK. MOPCR iK CeROYAe, COAe.
```

- K might be t, n , or s . Since I is t , we try $K \rightarrow n$.

```
DeQC VQeROC, i TirZ to RZOCe VCitivOY inBOCXQtioN CeWOCdinW OEeCQtioN XiDniWZt ReCEent. tZe
ReCEent ZQR QTQSenED QnD ReesSR itr ECeM. QYEZQ GPYP tZCee ZQR iDentiBieD Q CeQStZCOPWZ in
eneXM YOWiRtiVR. VOAcT QRRetR in ReVtOC CQAO nine ZOAe VonBiCXeD tZQt tZe RZiEXent OB
eNEenRiAe QCXOC TiYY EQRR tZCOPWZ COpte eVZO BiAe tOXOCCOT niWZt QR EeC tZe YoteRt
teYeXetCM DotQ. VOXXPniVQtioN TiYY e enVCMeteD TitZ VQeROC ViEZeC DPCinW tZe OEeCQtioN.
DeVCMet PRinW tZe ROXe SeM PEOk CeVeiet. i PCWe MOP to tQSe tZe PtXORT ECeVQPtioNR in
ZONdyinW tZiR XerRQWe QnD to enRPCe tZQt tZe inBOCXQtioN CeXQinR SnoTn OnYM to tCPRted
EeCROnneY. CeXeX eC Te QCe tZe WPOCDiOnR OB OPC nOtioNR ReVPCitM QnD OPC RPVVeRR ZinWeR On
ReVCeVM QnD ECeViRiOn. MOPCR in CeROYAe, COAe.
```

- $Z \rightarrow h$, forming “the” in the first line.
- “QCe” in line 8 might represent “are”.
- $D \rightarrow d$, $O \rightarrow o$
- “share” implies $R \rightarrow s$

```
dear Vaesar, i Tish to share VritiVaY inBorXation reWardinW oEeration XidniWht serEent. the
serEent has aTaSened and seeSs its EreM. aYEha GPYP three has identiBied a reaSthroPwh in
eneXM YowistiVs. VoAert assets in seVtor raAo nine haAe VonBirXed that the shiEXent OB
eNEensiAe arXor TiYY Eass throPwh roPte eVho BiAe toXorroT niWht as Eer the Yatest
teYeXetrM data. VoXXPniVation TiYY e enVrMETed Tith Vaesar ViEher dPrinW the oEeration.
deVrMET PsinW the saXe SeM PEOk reVeiet. i PrWe MoP to taSe the PtXost EreVaPtions in
handVinW this XessaWe and to ensPre that the inBorXation reXains SnoTn OnYM to trPsted
EersonneY. reXeX er Te are the WPardians OB oPr nations seVPritM and oPr sPVVess hinWes on
seVreVM and EreVision. MoPrs in resoYAe, raAe.
```

Additional substitutions:

$$\begin{aligned} V &\rightarrow c, & T &\rightarrow w, & Y &\rightarrow l, & B &\rightarrow f, \\ X &\rightarrow m, & W &\rightarrow g, & E &\rightarrow p, & S &\rightarrow k, \\ P &\rightarrow u, & M &\rightarrow y, & N &\rightarrow x, & A &\rightarrow v, & G &\rightarrow z \end{aligned}$$

```
dear caesar, i wish to share critical information regarding operation midnight serpent. the
serpent has awakened and seeks its prey. alpha zulu three has identified a reakthrough in
enemy logistics. covert assets in sector ravo nine have confirmed that the shipment of
expensive armor will pass through route echo five tomorrow night as per the latest
telemetry data. communication will e encrypted with caesar cipher during the operation.
decrypt using the same key upon receipt. i urge you to take the utmost precautions in
handling this message and to ensure that the information remains known only to trusted
personnel. remem er we are the guardians of our nations security and our success hinges on
secrecy and precision. yours in resolve, rave.
```

After all substitutions, the text becomes readable. Frequency analysis and substitution remain a trial-and-error method. Some errors may remain (for example, with the letter *b* accidentally replaced by space), but the general text is understandable.

Final Plaintext

“Dear Caesar, I wish to share critical information regarding Operation Midnight Serpent. The serpent has awakened and seeks its prey. Alpha Zulu Three has identified a breakthrough in enemy logistics. Covert assets in Sector Bravo Nine have confirmed that the shipment of expensive armor will pass through Route Echo Five tomorrow night as per the latest telemetry data. Communication will be encrypted with Caesar cipher during the operation. Decrypt using the same key upon receipt. I urge you to take the utmost precautions in handling this message and to ensure that the information remains known only to trusted personnel. Remember we are the guardians of our nation’s security and our success hinges on secrecy and precision. Yours in resolve, Brave.”